Math 54 — Week 4

Matrix operations; inverses; the invertible matrix theorem; determinants (Lay 2.1–2.3, 3.1)

§1 Facts used from Weeks 1–3

(W1) (Week 1) Row operations are reversible and do not change solution sets. Each matrix has exactly one reduced echelon form.

(W2) (Week 2) $A\mathbf{x} = x_1\mathbf{a}_1 + \cdots + x_n\mathbf{a}_n$, and $A(\mathbf{u}+\mathbf{v}) = A\mathbf{u} + A\mathbf{v}$, $A(c\mathbf{u}) = c(A\mathbf{u})$. $A\mathbf{x} = \mathbf{b}$ is consistent exactly when $\mathbf{b}$ is a combination of the columns.

(W3) (Weeks 2–3) For an $m \times n$ matrix A: a pivot in every row $\iff$ the columns span $\mathbb{R}^m$ $\iff$ $A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b}$ $\iff$ $\mathbf{x} \mapsto A\mathbf{x}$ is onto $\mathbb{R}^m$.

(W4) (Week 3) A pivot in every column $\iff$ the columns are linearly independent $\iff$ $A\mathbf{x} = \mathbf{0}$ has only the trivial solution $\iff$ $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.

(W5) (Week 3) A linear $T\colon \mathbb{R}^n \to \mathbb{R}^m$ has the standard matrix $[\,T(\mathbf{e}_1)\cdots T(\mathbf{e}_n)\,]$. Each pivot lies in a different row and column. More than n vectors in $\mathbb{R}^n$ are dependent, and a set containing $\mathbf{0}$ is dependent.

§2 This week

§2.1 Matrix operations (Lay 2.1)

The question. Let B rotate the plane by 90° counterclockwise and A reflect it through the x_1-axis. The move "first B, then A" sends $\mathbf{x}$ to $A(B\mathbf{x})$. Is it again given by a single matrix, and which one?

Definition. For an $m \times n$ matrix A, a_{ij} is the entry in row i and column j. The **diagonal entries** are $a_{11}, a_{22}, \ldots$. A square matrix whose entries off the diagonal are all 0 is a **diagonal matrix**; we write $\operatorname{diag}(d_1, \ldots, d_n)$ for the one with diagonal entries $d_1, \ldots, d_n$. The $m \times n$ **zero matrix**, written 0, has all entries 0. Two matrices are equal when they have the same size and the same entries.

Sums and scalar multiples. $A + B$ (same size) and rA are computed entry by entry. For matrices of one size and scalars r, s (Lay 2.1, Thm. 1):

$$A + B = B + A, \qquad (A + B) + C = A + (B + C), \qquad A + 0 = A,$$
$$r(A + B) = rA + rB, \qquad (r + s)A = rA + sA, \qquad r(sA) = (rs)A.$$

Each holds because it holds for numbers, applied entry by entry.

Definition. If A is $m \times n$ and $B = [\,\mathbf{b}_1 \cdots \mathbf{b}_p\,]$ is $n \times p$, the **product** AB is the $m \times p$ matrix

$$AB = [\,A\mathbf{b}_1 \;\; A\mathbf{b}_2 \;\; \cdots \;\; A\mathbf{b}_p\,].$$

The idea

The definition is chosen so that multiplying by AB does the same as multiplying by B and then by A: the product of two matrices is the matrix of the composite transformation.

Theorem (Products and composition)
For A $(m \times n)$, B $(n \times p)$ and $\mathbf{x}$ in $\mathbb{R}^p$, $(AB)\mathbf{x} = A(B\mathbf{x})$.

Proof. 1. By (W2), $B\mathbf{x} = x_1\mathbf{b}_1 + \cdots + x_p\mathbf{b}_p$.

2. By linearity of $\mathbf{y} \mapsto A\mathbf{y}$ (W2), $A(B\mathbf{x}) = x_1A\mathbf{b}_1 + \cdots + x_pA\mathbf{b}_p$.

3. By (W2) again, this is the matrix with columns $A\mathbf{b}_1, \ldots, A\mathbf{b}_p$ times $\mathbf{x}$, that is, $(AB)\mathbf{x}$.

□

Three ways to compute AB.

- **Sizes**: $(m \times n)(n \times p) = m \times p$; the inner sizes must agree, because $A\mathbf{b}_j$ needs $\mathbf{b}_j$ to have n entries.
- **Column view**: column j of AB is $A\mathbf{b}_j$, the combination of the columns of A with the entries of column j of B as weights (W2). So every column of AB lies in the span of the columns of A. Also, if $B\mathbf{c} = \mathbf{0}$ for some $\mathbf{c} \neq \mathbf{0}$, then $(AB)\mathbf{c} = A(B\mathbf{c}) = \mathbf{0}$: dependent columns of B give dependent columns of AB, with the same relation.
- **Row-column rule**: entry (i, j) of AB is entry i of $A\mathbf{b}_j$, which by the row-vector rule of Week 2 is

$$(AB)_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{in}b_{nj}.$$

  Example:
$$\begin{bmatrix} 3 & 1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} 1 & 4 & 0 \\ -2 & 1 & 5 \end{bmatrix} = \begin{bmatrix} 3-2 & 12+1 & 0+5 \\ -1-4 & -4+2 & 0+10 \end{bmatrix} = \begin{bmatrix} 1 & 13 & 5 \\ -5 & -2 & 10 \end{bmatrix}.$$

- **Row view**: row i of AB is (row i of A) B, because by the row-column rule both have entries $a_{i1}b_{1j} + \cdots + a_{in}b_{nj}$.

Theorem (Properties of matrix multiplication (Lay 2.1, Thm. 2))
For matrices of sizes for which the expressions are defined, and scalars r:

$$A(BC) = (AB)C, \qquad A(B+C) = AB + AC, \qquad (B+C)A = BA + CA,$$
$$r(AB) = (rA)B = A(rB), \qquad I_mA = A = AI_n.$$

Proof. 1. Associativity: for every $\mathbf{x}$, by the theorem on products and composition,

$$(A(BC))\mathbf{x} = A((BC)\mathbf{x}) = A(B(C\mathbf{x})), \qquad ((AB)C)\mathbf{x} = (AB)(C\mathbf{x}) = A(B(C\mathbf{x})).$$

Taking $\mathbf{x} = \mathbf{e}_j$ shows that column j is the same in both matrices.

2. $A(B+C) = AB + AC$: column j of the left side is $A(\mathbf{b}_j + \mathbf{c}_j) = A\mathbf{b}_j + A\mathbf{c}_j$ by linearity. $(B+C)A = BA + CA$: by the row-column rule, entry (i, j) is $\sum_k (b_{ik} + c_{ik})a_{kj} = \sum_k b_{ik}a_{kj} + \sum_k c_{ik}a_{kj}$.

3. $r(AB) = (rA)B = A(rB)$: column j is $r(A\mathbf{b}_j) = (rA)\mathbf{b}_j = A(r\mathbf{b}_j)$, by linearity and the definition of $A\mathbf{x}$.

4. $I_mA = A$ because column j is $I_m\mathbf{a}_j = \mathbf{a}_j$; $AI_n = A$ because column j is $A\mathbf{e}_j = \mathbf{a}_j$.

□

Powers. For square A, $A^k = A \cdots A$ (k factors) and $A^0 = I$.

Diagonal matrices scale. For $D = \operatorname{diag}(d_1, \ldots, d_n)$, AD multiplies column j of A by d_j, and DA multiplies row i of A by d_i. Reason: column j of AD is $A(d_j \mathbf{e}_j) = d_j \mathbf{a}_j$; row i of DA is (row i of D)A $= d_i$ (row i of A). Instance:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 2 & 10 \\ 6 & 20 \end{bmatrix}, \qquad \begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 15 & 20 \end{bmatrix}.$$

Three warnings.

(1) $AB \neq BA$ in general. For example

$$C = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}, \quad D = \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}: \qquad CD = \begin{bmatrix} 7 & 2 \\ 3 & 1 \end{bmatrix}, \quad DC = \begin{bmatrix} 1 & 2 \\ 3 & 7 \end{bmatrix}.$$

(2) $AB = AC$ does not imply $B = C$: with $A = \operatorname{diag}(1, 0)$, $B = I$ and $C = \operatorname{diag}(1, 2)$, both products equal A (the diagonal rule), but $B \neq C$.

(3) $AB = 0$ does not imply $A = 0$ or $B = 0$:

$$\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Definition. The **transpose** A^T of an $m \times n$ matrix A is the $n \times m$ matrix whose columns are the rows of A: $(A^T)_{ij} = a_{ji}$.

Theorem (Transposes (Lay 2.1, Thm. 3))
$(A^T)^T = A$, $(A + B)^T = A^T + B^T$, $(rA)^T = rA^T$, and $(AB)^T = B^T A^T$.

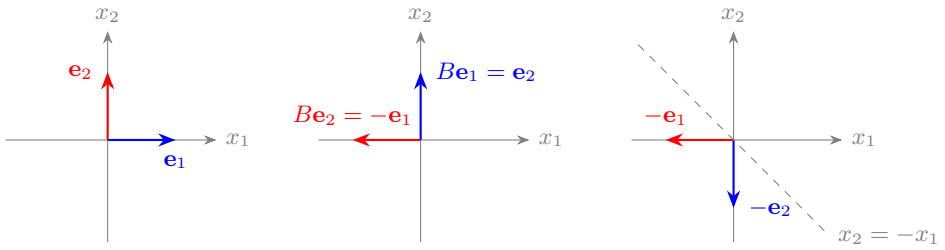
Proof. The first three hold entry by entry. For the product, by the row-column rule,

$$\big((AB)^T\big)_{ij} = (AB)_{ji} = \sum_k a_{jk} b_{ki} = \sum_k (B^T)_{ik} (A^T)_{kj} = (B^T A^T)_{ij}.$$

□

Inner and outer products. For $\mathbf{u}, \mathbf{v}$ in $\mathbb{R}^n$ (as $n \times 1$ matrices), $\mathbf{u}^T \mathbf{v}$ is 1×1, with entry $u_1 v_1 + \cdots + u_n v_n$ (the **inner product**), and $\mathbf{u}\mathbf{v}^T$ is $n \times n$, with entries $u_i v_j$ (the **outer product**). Instance, with $\mathbf{u} = (1, 2)$ and $\mathbf{v} = (3, -1)$:

$$\mathbf{u}^T \mathbf{v} = \begin{bmatrix} 1 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \end{bmatrix}, \qquad \mathbf{u}\mathbf{v}^T = \begin{bmatrix} 1 \\ 2 \end{bmatrix} \begin{bmatrix} 3 & -1 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ 6 & -2 \end{bmatrix}.$$



$\mathbf{e}_1, \mathbf{e}_2$ after B (rotate 90°) after A: $AB\mathbf{e}_1$, $AB\mathbf{e}_2$

The question. B sends $\mathbf{e}_1 \mapsto \mathbf{e}_2$ and $\mathbf{e}_2 \mapsto -\mathbf{e}_1$; then A (reflection through the x_1-axis) sends $\mathbf{e}_2 \mapsto -\mathbf{e}_2$ and $-\mathbf{e}_1 \mapsto -\mathbf{e}_1$. The final pair is the reflection of $\mathbf{e}_1, \mathbf{e}_2$ through the line $x_2 = -x_1$.

The question answered. Yes: by the theorem on products and composition, "first B, then A" is $\mathbf{x} \mapsto (AB)\mathbf{x}$. With

$$B = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}, \qquad A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}, \qquad AB = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix},$$

which is the reflection through the line $x_2 = -x_1$: it sends $\mathbf{e}_1$ to $-\mathbf{e}_2$ and $\mathbf{e}_2$ to $-\mathbf{e}_1$ (figure).

Shortcuts

- **Size check before multiplying**: $(m \times n)(n \times p)$ is $m \times p$; if the inner numbers differ, the product is undefined. Reason: the definition. Instance: $(2 \times 2)(2 \times 3)$ is 2×3, as in the row-column example.
- **One entry of AB**: row i of A times column j of B; the rest of the product is not needed. Reason: the row-column rule. Instance: entry $(2, 3)$ of the row-column example is $(-1) \cdot 0 + 2 \cdot 5 = 10$.
- **Compose geometric moves by tracking $\mathbf{e}_1, \mathbf{e}_2$**: the columns of AB are $A(B\mathbf{e}_1)$ and $A(B\mathbf{e}_2)$. Reason: $(AB)\mathbf{x} = A(B\mathbf{x})$. Instance: the question.
- **Diagonal factors**: on the right they scale columns, on the left they scale rows. Reason and instance: above.

Traps

- $AB \neq BA$ in general: $CD \neq DC$ above.
- $(A+B)^2 = A^2 + AB + BA + B^2$, not $A^2 + 2AB + B^2$; the two differ by $BA - AB$, which is not 0 for C, D above.
- No cancelling: $AB = AC$ with $B \neq C$ in warning (2).

§2.2 The inverse of a matrix (Lay 2.2)

The question. The system $3x_1 + 4x_2 = 3$, $5x_1 + 6x_2 = 7$ is $A\mathbf{x} = \mathbf{b}$ with the matrix below. For numbers, $ax = b$ is solved by $x = a^{-1}b$ when $a \neq 0$. Is there a matrix A^{-1} with $\mathbf{x} = A^{-1}\mathbf{b}$, when does one exist, and how do we compute it?

$$A = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Definition. An $n \times n$ matrix A is **invertible** if there is an $n \times n$ matrix C with $CA = I$ and $AC = I$. A matrix that is not invertible is **singular**.

Such a C is unique: if B also has $BA = I$ and $AB = I$, then

$$B = BI = B(AC) = (BA)C = IC = C.$$

It is written A^{-1}.

Theorem (The 2×2 inverse (Lay 2.2, Thm. 4))
The matrix below is invertible if and only if $ad - bc \neq 0$, and then A^{-1} is as shown. The number $ad - bc$ is the **determinant** $\det A$.

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, \qquad A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Proof. 1. Let M be the matrix below. By the row-column rule, in both orders,

$$M = \begin{bmatrix} d & -b \\ -c & a \end{bmatrix},$$

$$AM = \begin{bmatrix} ad - bc & -ab + ba \\ cd - dc & -cb + da \end{bmatrix} = (ad - bc)I, \qquad MA = \begin{bmatrix} da - bc & db - bd \\ -ca + ac & -cb + ad \end{bmatrix} = (ad - bc)I.$$

2. **Assume $ad - bc \neq 0$.** Dividing by $ad - bc$ shows that $\frac{1}{ad-bc}M$ is a two-sided inverse.
3. **Assume $ad - bc = 0$.** Then $AM = 0$, so $A\mathbf{x} = \mathbf{0}$ for each column $\mathbf{x}$ of M. If an invertible A had $A\mathbf{x} = \mathbf{0}$, then $\mathbf{x} = A^{-1}(A\mathbf{x}) = A^{-1}\mathbf{0} = \mathbf{0}$; so it is enough to find a nonzero column of M. If $A \neq 0$, some entry of A is nonzero,

and the entries of M are those of A up to sign and position, so some column of M is nonzero. If $A = 0$, then $CA = 0 \neq I$ for every C. Either way A is not invertible.

□

Theorem (Invertible matrices give unique solutions (Lay 2.2, Thm. 5))
If A is an invertible $n \times n$ matrix, then for each $\mathbf{b}$ in $\mathbb{R}^n$ the equation $A\mathbf{x} = \mathbf{b}$ has the unique solution $\mathbf{x} = A^{-1}\mathbf{b}$.

Proof. 1. $A(A^{-1}\mathbf{b}) = (AA^{-1})\mathbf{b} = I\mathbf{b} = \mathbf{b}$, so $A^{-1}\mathbf{b}$ is a solution.

2. If $A\mathbf{u} = \mathbf{b}$, multiply on the left by A^{-1}: $\mathbf{u} = (A^{-1}A)\mathbf{u} = A^{-1}\mathbf{b}$. So it is the only solution.

□

The question answered (formula). $\det A = 18 - 20 = -2 \neq 0$, so

$$A^{-1} = \frac{1}{-2}\begin{bmatrix} 6 & -4 \\ -5 & 3 \end{bmatrix} = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix}, \qquad \mathbf{x} = A^{-1}\mathbf{b} = \begin{bmatrix} -3 & 2 \\ 5/2 & -3/2 \end{bmatrix}\begin{bmatrix} 3 \\ 7 \end{bmatrix} = \begin{bmatrix} -9+14 \\ 15/2 - 21/2 \end{bmatrix} = \begin{bmatrix} 5 \\ -3 \end{bmatrix}.$$

Check: $3 \cdot 5 + 4 \cdot (-3) = 3$ and $5 \cdot 5 + 6 \cdot (-3) = 7$.

Columns of A^{-1}. Column j of $AA^{-1} = I$ is A(column j of A^{-1}) $= \mathbf{e}_j$. So column j of A^{-1} is the solution of $A\mathbf{x} = \mathbf{e}_j$.

Theorem (Algebra of inverses (Lay 2.2, Thm. 6))
If A and B are invertible $n \times n$ matrices, then: (a) A^{-1} is invertible and $(A^{-1})^{-1} = A$; (b) AB is invertible and $(AB)^{-1} = B^{-1}A^{-1}$; (c) A^T is invertible and $(A^T)^{-1} = (A^{-1})^T$.

Proof. 1. (a) The equations $A^{-1}A = I$ and $AA^{-1} = I$ say that A is a two-sided inverse of A^{-1}.

2. (b) By associativity, in both orders,

$$(AB)(B^{-1}A^{-1}) = A(BB^{-1})A^{-1} = AA^{-1} = I, \qquad (B^{-1}A^{-1})(AB) = B^{-1}(A^{-1}A)B = B^{-1}B = I.$$

3. (c) By $(XY)^T = Y^T X^T$, in both orders,

$$A^T(A^{-1})^T = (A^{-1}A)^T = I^T = I, \qquad (A^{-1})^T A^T = (AA^{-1})^T = I^T = I.$$

□

The order reverses in (b) because AB means "first B, then A"; to undo it, undo A first and then B. Applying (b) repeatedly, a product of invertible $n \times n$ matrices is invertible, and its inverse is the product of the inverses in the reverse order.

Solving matrix equations. If A is invertible, multiplying $AX = B$ on the left by A^{-1} gives $X = A^{-1}B$. In the same way, if A and B are invertible, $AXB = C$ gives $X = A^{-1}CB^{-1}$ (multiply on the left by A^{-1} and on the right by B^{-1}), and $AB = AC$ gives $B = C$. To find $A^{-1}B$ without A^{-1}, row reduce $[\,A \;\; B\,]$: the operations that turn A into I solve $A\mathbf{x} = \mathbf{b}_j$ for every column $\mathbf{b}_j$ at once, so the result is $[\,I \;\; A^{-1}B\,]$.

Elementary matrices. An **elementary matrix** E is obtained by doing one row operation to I_m.

- **EA is the result of doing that row operation to A.** Reason: row i of EA is (row i of E)A (row view). A row of E that the operation did not change is $\mathbf{e}_k^T$, and $\mathbf{e}_k^T A$ is row k of A. For the replacement $R_i + cR_j$, row i of E is $\mathbf{e}_i^T + c\mathbf{e}_j^T$, giving (row i of A)$+c$(row j of A). Interchange and scaling are checked the same way.
- **Every elementary matrix is invertible, and its inverse is the elementary matrix F of the reverse operation.** $FE = F(EI)$ is I after the operation and then its reverse, which is I. In the same way $EF = I$.

A square matrix with n pivots has reduced echelon form I_n. In the RREF of an $n \times n$ matrix with n pivots, every row has a leading 1, and each is strictly to the right of the one above. Row 1's is in column 1 or later, row 2's in column 2 or later, and so on; row n's is in column n or later, hence in column n. So the leading 1 of row i is in column i for every i (otherwise row n would need a column past n). Every column is a pivot column, so its other entries are 0: the RREF is I_n.

Theorem (Invertibility and row reduction (Lay 2.2, Thm. 7))

An $n \times n$ matrix A is invertible if and only if A is row equivalent to I_n. In that case, any sequence of row operations that reduces A to I_n also transforms I_n into A^{-1}.

The idea

Reducing A to I by row operations means multiplying on the left by elementary matrices: $E_k \cdots E_1 A = I$. So the product $E_k \cdots E_1$ undoes A, and it is what the same row operations do to I. Writing A and I side by side as $[\,A \;\; I\,]$ does both at once.

Proof. 1. **Assume A is invertible.** By Theorem 5, $A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b}$, so A has a pivot in every row (W3): n pivots. By the paragraph above, the RREF of A is I_n.

2. **Assume A is row equivalent to I_n:** $E_k \cdots E_1 A = I$ for elementary E_i. Let $C = E_k \cdots E_1$; it is invertible, as a product of invertible matrices. Then $A = C^{-1}(CA) = C^{-1}$, so A is invertible (Theorem 6(a)) and $A^{-1} = C$.

3. $C = CI = E_k \cdots E_1 I$ is the result of applying the same row operations to I.

□

Computing A^{-1}. Row reduce $[\,A \;\; I\,]$. If A reduces to I, the result is $[\,I \;\; A^{-1}\,]$. If A has fewer than n pivots (for example, a zero row appears in the left half), A is not invertible.

Worked example. Find the inverse of the matrix A on the left of $[\,A \;\; I\,]$:

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 2 & 1 & 5 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 1 & 2 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_3 - R_2} \left[\begin{array}{ccc|ccc} 1 & 0 & 2 & 1 & 0 & 0 \\ 0 & 1 & 1 & -2 & 1 & 0 \\ 0 & 0 & 1 & 2 & -1 & 1 \end{array}\right]$$

$$\xrightarrow[R_1 - 2R_3]{R_2 - R_3} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -3 & 2 & -2 \\ 0 & 1 & 0 & -4 & 2 & -1 \\ 0 & 0 & 1 & 2 & -1 & 1 \end{array}\right], \qquad A^{-1} = \begin{bmatrix} -3 & 2 & -2 \\ -4 & 2 & -1 \\ 2 & -1 & 1 \end{bmatrix}.$$

Check column 1 of A^{-1}, which must solve $A\mathbf{x} = \mathbf{e}_1$:

$$\begin{bmatrix} 1 & 0 & 2 \\ 2 & 1 & 5 \\ 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} -3 \\ -4 \\ 2 \end{bmatrix} = \begin{bmatrix} -3 + 0 + 4 \\ -6 - 4 + 10 \\ 0 - 4 + 4 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}.$$

The question answered. A^{-1} exists exactly when A reduces to I; for 2×2 matrices, exactly when $ad - bc \neq 0$. Then $\mathbf{x} = A^{-1}\mathbf{b}$, and A^{-1} is computed by the formula (2×2) or by row reducing $[\,A \;\; I\,]$. For the question's matrix this gave $\mathbf{x} = (5, -3)$.

Shortcuts

- **2×2 inverse**: swap the diagonal entries, negate the off-diagonal ones, divide by $ad - bc$. Condition: $ad - bc \neq 0$. Reason: Theorem 4. Instance: the question.
- **Several systems with the same A**: row reduce $[\,A \;\; \mathbf{b}_1 \;\; \mathbf{b}_2\,]$ once. Reason: the same row operations reduce A in each augmented matrix. Instance: one replacement handles both right-hand sides,

  $$\left[\begin{array}{cc|cc} 1 & 2 & 3 & 1 \\ 3 & 5 & 8 & 1 \end{array}\right] \xrightarrow{R_2 - 3R_1} \left[\begin{array}{cc|cc} 1 & 2 & 3 & 1 \\ 0 & -1 & -1 & -2 \end{array}\right],$$

  so $x_2 = 1$, $x_1 = 1$ for the first system and $x_2 = 2$, $x_1 = -3$ for the second: the solutions are $(1, 1)$ and $(-3, 2)$.
- **One column of A^{-1}**: solve $A\mathbf{x} = \mathbf{e}_j$. Reason: the paragraph on columns of A^{-1}. Instance: the check after the worked example.
- **Matrix equations**: multiply by inverses on the side where the factor sits. Reason: $A^{-1}A = I$ removes A only when A^{-1} is next to it. Instance: $AXB = C$ gives $X = A^{-1}CB^{-1}$.

Traps

- $(CD)^{-1} = D^{-1}C^{-1}$, not $C^{-1}D^{-1}$. For C, D in 2.1, the two products differ:

$$C^{-1} = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}, \quad D^{-1} = \begin{bmatrix} 1 & 0 \\ -3 & 1 \end{bmatrix}, \qquad D^{-1}C^{-1} = \begin{bmatrix} 1 & -2 \\ -3 & 7 \end{bmatrix}, \quad C^{-1}D^{-1} = \begin{bmatrix} 7 & -2 \\ -3 & 1 \end{bmatrix}.$$

- From $AX = B$ the solution is $X = A^{-1}B$, not BA^{-1}. With $A = C$ and $B = D$ from 2.1, the two products differ:

$$C^{-1}D = \begin{bmatrix} -5 & -2 \\ 3 & 1 \end{bmatrix}, \qquad DC^{-1} = \begin{bmatrix} 1 & -2 \\ 3 & -5 \end{bmatrix}.$$

- In $[\,A \;\; I\,]$, apply every row operation to the whole row, both halves; otherwise the right half is not A^{-1}.
- Only square matrices have inverses. For the matrices below, $AD = I_2$. But $A\mathbf{e}_3 = \mathbf{0}$, so for every 3×2 matrix C

$$CA\mathbf{e}_3 = C(A\mathbf{e}_3) = C\mathbf{0} = \mathbf{0} \neq \mathbf{e}_3,$$

and $CA \neq I_3$.

$$A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}, \qquad D = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

§2.3 The invertible matrix theorem (Lay 2.3)

The question. The square matrix below has dependent columns. Does that already tell us whether $A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b}$, and whether A is invertible?

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Theorem (The invertible matrix theorem (Lay 2.3, Thm. 8))

For a *square* $n \times n$ matrix A, the following are equivalent (all true or all false):

(a) A is invertible.
(b) A is row equivalent to I_n.
(c) A has n pivot positions.
(d) $A\mathbf{x} = \mathbf{0}$ has only the trivial solution.
(e) The columns of A are linearly independent.
(f) $\mathbf{x} \mapsto A\mathbf{x}$ is one-to-one.
(g) $A\mathbf{x} = \mathbf{b}$ has at least one solution for each $\mathbf{b}$.
(h) The columns of A span $\mathbb{R}^n$.
(i) $\mathbf{x} \mapsto A\mathbf{x}$ maps $\mathbb{R}^n$ onto $\mathbb{R}^n$.
(j) There is C with $CA = I$.
(k) There is D with $AD = I$.
(l) A^T is invertible.

The idea

For a square matrix, "a pivot in every row" and "a pivot in every column" both say "n pivots". So the list (g)–(i) from (W3) and the list (d)–(f) from (W4) describe the same matrices, and (b), (c) are two more ways to say n pivots. A one-sided inverse forces one of these lists, and hence a two-sided inverse.

Proof. 1. (a) $\iff$ (b) is Theorem 7. (c)$\Rightarrow$(b) is the paragraph before Theorem 7. (b)$\Rightarrow$(c): the RREF of A is I_n (W1), which has n pivots.

2. (c) $\iff$ (d) $\iff$ (e) $\iff$ (f): each of the n columns holds at most one pivot, so n pivots means exactly a pivot in every column; then use (W4).

3. (c) $\iff$ (g) $\iff$ (h) $\iff$ (i): each of the n rows holds at most one pivot, so n pivots means exactly a pivot in every row; then use (W3).

4. (a)⇒(j) and (a)⇒(k): take $C = D = A^{-1}$.
5. (j)⇒(d): if $CA = I$ and $A\mathbf{x} = \mathbf{0}$, then
$$\mathbf{x} = (CA)\mathbf{x} = C(A\mathbf{x}) = C\mathbf{0} = \mathbf{0}.$$
6. (k)⇒(g): if $AD = I$, then for each $\mathbf{b}$, $\mathbf{x} = D\mathbf{b}$ solves $A\mathbf{x} = \mathbf{b}$, because $A(D\mathbf{b}) = (AD)\mathbf{b} = \mathbf{b}$.
7. (a)⇒(l) is Theorem 6(c). (l)⇒(a): apply Theorem 6(c) to A^T, and use $(A^T)^T = A$.

□

Consequences.

- **A one-sided inverse is two-sided.** If A, B are $n \times n$ and $AB = I$, then A is invertible by (k), $B = A^{-1}(AB) = A^{-1}$, and $BA = I$.
- **Factors of an invertible product are invertible.** If A, B are $n \times n$ and AB is invertible with inverse W, then $A(BW) = I$, so A is invertible by (k); and $(WA)B = I$, so B is invertible by (j).
- **Triangular matrices.** A square triangular matrix is invertible exactly when all its diagonal entries are nonzero.
  1. **Assume A is upper triangular with nonzero diagonal entries.** It is already an echelon form, with a pivot on each diagonal position: n pivots, so (c) holds.
  2. **Assume A is upper triangular and some $a_{kk} = 0$,** with k the first such index. Columns $1, \ldots, k$ have zero entries in rows $k, \ldots, n$. If $k = 1$, column 1 is $\mathbf{0}$, so the columns are dependent (W5). If $k > 1$, deleting those zero rows turns the first k columns into k vectors in $\mathbb{R}^{k-1}$, which are dependent (W5); the same relation holds for the columns of A. Either way (e) fails.
  3. A lower triangular A has an upper triangular transpose with the same diagonal; use (l).

Invertible linear transformations. $T\colon \mathbb{R}^n \to \mathbb{R}^n$ is **invertible** if there is $S\colon \mathbb{R}^n \to \mathbb{R}^n$ with $S(T(\mathbf{x})) = \mathbf{x}$ and $T(S(\mathbf{x})) = \mathbf{x}$ for all $\mathbf{x}$; then S is written T^{-1}.

Theorem (Invertible linear transformations (Lay 2.3, Thm. 9))
Let $T\colon \mathbb{R}^n \to \mathbb{R}^n$ be linear with standard matrix A. Then T is invertible if and only if A is invertible, and then $T^{-1}(\mathbf{x}) = A^{-1}\mathbf{x}$.

Proof. 1. **Assume A is invertible.** Let $S(\mathbf{x}) = A^{-1}\mathbf{x}$. Then $S(T(\mathbf{x})) = A^{-1}A\mathbf{x} = \mathbf{x}$ and $T(S(\mathbf{x})) = AA^{-1}\mathbf{x} = \mathbf{x}$.

2. **Assume T is invertible, with S as in the definition.** Every $\mathbf{b}$ equals $T(S(\mathbf{b}))$, so T is onto $\mathbb{R}^n$, and by (i)⇒(a) A is invertible.
3. Then for each $\mathbf{x}$, $\mathbf{x} = T(A^{-1}\mathbf{x})$, so $S(\mathbf{x}) = S(T(A^{-1}\mathbf{x})) = A^{-1}\mathbf{x}$.

□

Worked example: a formula for T^{-1}. Let $T(x_1, x_2) = (2x_1 + x_2,\ 5x_1 + 3x_2)$. Its standard matrix and inverse are
$$A = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}, \qquad A^{-1} = \frac{1}{6-5}\begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -5 & 2 \end{bmatrix},$$
so $T^{-1}(x_1, x_2) = (3x_1 - x_2,\ -5x_1 + 2x_2)$. Check: T applied to this gives first entry $2(3x_1 - x_2) + (-5x_1 + 2x_2) = x_1$ and second entry $5(3x_1 - x_2) + 3(-5x_1 + 2x_2) = x_2$.

The question answered. Yes. Statement (e) fails, so every statement fails: A is not invertible, and $A\mathbf{x} = \mathbf{b}$ has no solution for some $\mathbf{b}$. To exhibit one, reduce with $\mathbf{b}$ as letters:
$$\left[\begin{array}{cc|c} 1 & 2 & b_1 \\ 2 & 4 & b_2 \end{array}\right] \xrightarrow{R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 2 & b_1 \\ 0 & 0 & b_2 - 2b_1 \end{array}\right].$$
For $\mathbf{b} = (1, 0)$ the last row is $0 = -2$.

Shortcuts

- **Invertible or not?** For a square matrix, any one IMT statement settles it: count pivots, spot two equal or proportional columns or rows, a zero row or column, or compute $ad - bc$ for a 2×2. Reason: the theorem. Instance: the question (proportional columns).
- **One-sided inverse suffices**: for square A, B, $AB = I$ already gives $BA = I$ and $B = A^{-1}$. Reason: the first consequence. Instance: for A and A^{-1} in the T^{-1} example, checking $AA^{-1} = I$ is enough.
- **Triangular**: invertible exactly when no diagonal entry is 0. Reason: the third consequence. Instance: the upper triangular matrix below is singular.
$$\begin{bmatrix} 2 & 5 \\ 0 & 0 \end{bmatrix}$$
- **Square maps**: $T\colon \mathbb{R}^n \to \mathbb{R}^n$ is one-to-one exactly when it is onto. Reason: (f) $\iff$ (i). Instance: the rotation of Week 3 is one-to-one, hence onto.

Traps

- The theorem is only for square matrices. The 2×3 matrix below has a pivot in every row, but its columns are dependent (three vectors in $\mathbb{R}^2$):
$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$
- For non-square matrices a one-sided inverse says nothing about the other side: the A and D in the last trap of 2.2 have $AD = I_2$ but $DA \neq I_3$.
- "$A\mathbf{x} = \mathbf{b}$ is solvable for every $\mathbf{b}$" fails for every singular square matrix, even when it holds for some $\mathbf{b}$: in the question, $\mathbf{b} = (1, 2)$ works but $(1, 0)$ does not.

§2.4 Introduction to determinants (Lay 3.1)

The question. For 2×2 matrices, the number $ad - bc$ decides invertibility. Is there such a number for $n \times n$ matrices, and how do we compute it quickly for the matrix below?

$$A = \begin{bmatrix} 1 & 5 & 0 \\ 2 & 4 & -1 \\ 0 & -2 & 0 \end{bmatrix}$$

Definition. For $n \geq 2$, let A_{ij} be the $(n-1) \times (n-1)$ matrix obtained by deleting row i and column j of A. The (i, j)-**cofactor** is $C_{ij} = (-1)^{i+j} \det A_{ij}$. The **determinant** is defined recursively by expanding along the first row,

$$\det A = a_{11}C_{11} + a_{12}C_{12} + \cdots + a_{1n}C_{1n},$$

with $\det[\,a\,] = a$ for a 1×1 matrix. We also write $|A|$ for $\det A$.

For $n = 2$ this gives $a_{11}a_{22} - a_{12}a_{21} = ad - bc$. The signs $(-1)^{i+j}$ alternate in a checkerboard pattern, with $+$ in the top left:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

Theorem (Cofactor expansion (Lay 3.1, Thm. 1))

$\det A$ can be computed by expanding along any row i or down any column j:

$$\det A = a_{i1}C_{i1} + \cdots + a_{in}C_{in} = a_{1j}C_{1j} + \cdots + a_{nj}C_{nj}.$$

Lay states this theorem without proof; it is cited here. Its use is speed: a zero entry a_{ij} makes the term $a_{ij}C_{ij}$ zero, so that cofactor never has to be computed.

Theorem (Triangular matrices (Lay 3.1, Thm. 2))

If A is triangular, $\det A$ is the product of the diagonal entries of A.

Proof. By induction on n.

1. For $n = 1$, $\det[\,a\,] = a$.
2. Assume the theorem for $(n-1) \times (n-1)$ matrices, and let A be upper triangular $n \times n$. In column 1 only a_{11} can be nonzero, so expanding down column 1 gives $\det A = a_{11}C_{11} = a_{11} \det A_{11}$ (the sign of $(1,1)$ is $+$).
3. A_{11} is upper triangular with diagonal entries $a_{22}, \ldots, a_{nn}$, so by the assumption $\det A_{11} = a_{22} \cdots a_{nn}$, and $\det A = a_{11}a_{22} \cdots a_{nn}$.
4. For lower triangular A, expand along row 1 instead: only a_{11} can be nonzero, and A_{11} is lower triangular.

□

Example:

$$\det \begin{bmatrix} 2 & 0 & 0 \\ 7 & -3 & 0 \\ 5 & 1 & 4 \end{bmatrix} = 2 \cdot (-3) \cdot 4 = -24.$$

The question answered. Column 3 has one nonzero entry, $a_{23} = -1$, with sign $(-1)^{2+3} = -1$. Expanding down column 3,

$$\det A = a_{23}C_{23} = (-1) \cdot (-1) \det \begin{bmatrix} 1 & 5 \\ 0 & -2 \end{bmatrix} = 1 \cdot (-2 - 0) = -2.$$

Expanding along row 1 instead gives the same number, as Theorem 1 says:

$$\det A = 1 \det \begin{bmatrix} 4 & -1 \\ -2 & 0 \end{bmatrix} - 5 \det \begin{bmatrix} 2 & -1 \\ 0 & 0 \end{bmatrix} + 0 = 1 \cdot (0 - 2) - 5 \cdot 0 = -2.$$

For 2×2 matrices the determinant decides invertibility (Theorem 4); that the same holds for $n \times n$ matrices is proved in Week 5.

Shortcuts

- **Expand along the row or column with the most zeros**, minding the checkerboard signs. Reason: a zero entry removes its cofactor from the sum (Theorem 1). Instance: column 3 in the question.
- **Repeat on the minors**: after one expansion, expand each smaller determinant along its own best row or column. Instance: P6.
- **Triangular**: multiply the diagonal. Reason: Theorem 2. Instance: -24 above.

Traps

- Cofactor signs: $C_{ij} = (-1)^{i+j} \det A_{ij}$. Dropping the sign in the question gives $+2$ instead of -2.
- "Multiply the diagonal" is only for triangular matrices: the matrix below has diagonal product 0, but its determinant is $0 \cdot 0 - 1 \cdot 1 = -1$:

$$\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

- $|A|$ means $\det A$, not an absolute value: it can be negative, as $|A| = -2$ in the question shows.

§3 Homework this week

HW3, Sections 2.1–2.3, and HW4, Section 3.1, with a one-line hint for each exercise. (The rest of HW4 is in the Week 5 handout.)

Section 2.1.

- **#1, #3** (combinations and products). Check sizes first; $(3I)A = 3A$.
- **#7** (size of B). $(m \times n)(n \times p) = m \times p$.
- **#9** (k with $AB = BA$). Compute both products with k as a letter and compare entries.
- **#11** (AD and DA for diagonal D; a B other than I and 0 with $AB = BA$). A diagonal factor on the right scales the columns, on the left the rows; every matrix commutes with each scalar multiple of I.
- **#12** ($B \neq 0$ with $AB = 0$). Column view: each column of B is a solution of $A\mathbf{x} = \mathbf{0}$; find one nontrivial solution.
- **#15, #17** (true/false: columns of AB). Definition of AB: column j is $A\mathbf{b}_j$, a combination of the columns of A with weights from $\mathbf{b}_j$.
- **#22** (true/false: $(AB)^T$). Theorem 3(d): the transpose of a product reverses the order.
- **#35** ($\mathbf{u}^T\mathbf{v}$, $\mathbf{v}^T\mathbf{u}$, $\mathbf{u}\mathbf{v}^T$, $\mathbf{v}\mathbf{u}^T$). Inner and outer products: a 1×1 matrix and a 3×3 matrix; compare each pair by transposing.

Section 2.2.

- **#1, #5, #7** (2×2 inverse; check it; solve with it). Theorem 4, multiply to check, then Theorem 5.
- **#9** (four systems with one A). Row reduce $[\,A \;\; \mathbf{b}_1 \;\; \cdots \;\; \mathbf{b}_4\,]$ once.
- **#12, #17, #18, #19, #20** (true/false). Theorem 6(b), Theorem 5, Theorem 7, inverses of elementary matrices, and the proof of Theorem 7.
- **#23** (cancel A). Multiply on the left by A^{-1}; the warning in 2.1 shows why invertibility is needed.
- **#29** (solve a matrix equation for X). Multiply by inverses on the correct sides, keeping the order (solving matrix equations).
- **#31, #33** (invertible $\Rightarrow$ independent columns; only the trivial solution $\Rightarrow$ invertible). Theorem 5 with $\mathbf{b} = \mathbf{0}$; n pivots and Theorem 7.
- **#41** (3×3 inverse). Row reduce $[\,A \;\; I\,]$ (worked example).

Section 2.3.

- **#3, #7, #21** (invertible?). Count pivots; for triangular matrices check the diagonal (third consequence).
- **#11, #13, #15** (true/false). The IMT, which needs a square matrix; for #15 use a singular square matrix.
- **#23** (two identical columns). Dependent columns; statement (e).
- **#28, #33** ($EF = I$; verify the boxed statement). A one-sided inverse is two-sided (first consequence).
- **#38** (one-to-one square map). (f) $\Longleftrightarrow$ (i) $\Longleftrightarrow$ (a).
- **#41, #44** (inverse of a linear map; onto implies invertible). Theorem 9 and the T^{-1} worked example; for #44 use (i)$\Rightarrow$(a).

Section 3.1.

- **#1, #5** (3×3 determinants). Expand along row 1, and (for #1) also down column 2; the answers must agree (Theorem 1).
- **#11** (triangular). Theorem 2.
- **#13** (5×5). Expand along the row or column with the most zeros, and repeat on each minor.
- **#21** (a row replacement in a 2×2 matrix). Compute both determinants with the letters in place and compare.

§4 Problem set

P1. For the matrices below, compute AB, BA, $(AB)^T$ and B^TA^T.

$$A = \begin{bmatrix} 2 & -1 \\ 1 & 3 \end{bmatrix}, \qquad B = \begin{bmatrix} 0 & 1 \\ 4 & -2 \end{bmatrix}$$

P2. Compute X^3 for the matrix below, guess X^k, and prove the guess.

$$X = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$$

P3. Use the 2×2 inverse to solve $2x_1 + 5x_2 = 1$, $x_1 + 3x_2 = -1$, and then the system with the same coefficients and right-hand side $(0, 1)$.

P4. Find A^{-1} for the matrix below, and check your answer.

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

P5. Is the matrix below invertible? Answer with as little computation as possible.

$$\begin{bmatrix} 1 & -2 & 3 \\ 2 & -4 & 6 \\ 0 & 1 & 5 \end{bmatrix}$$

P6. Compute the determinant below by choosing good rows and columns.

$$\det \begin{bmatrix} 3 & 0 & 4 & 1 \\ 2 & 0 & -1 & 0 \\ 5 & 7 & 3 & 2 \\ 1 & 0 & 2 & 0 \end{bmatrix}$$

P7. True or false, with a reason (all matrices $n \times n$). (a) If $A\mathbf{x} = \mathbf{0}$ has a nontrivial solution, then A has fewer than n pivots. (b) If the columns of A span $\mathbb{R}^n$, then they are linearly independent. (c) If $AB = I$, then A and B are both invertible. (d) If A is invertible, then A^2 is invertible.

P8. Suppose A is $n \times n$ and $A^3 = 0$. Show that $I - A$ is invertible, with inverse $I + A + A^2$.